SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH314

Examination - January Semester 2023

Real Analysis II

Tuesday, 23 May 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **SEVEN (7)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Total 100 marks)

Question 1a

Define a function

$$g(x) = \begin{cases} 7x^5 + 3x + 4\cos(\pi x), & x \geq 0 \\ 3\sin(x) + 4, & x < 0 \end{cases}.$$

Question 1a(i)

Show that g is continuous at $x = 0$ and show that g is differentiable at $x = 0$ by computing the one-sided limits

$$\lim_{x \longrightarrow 0^-} g'(x) \text{ and } \lim_{x \longrightarrow 0^+} g'(x).$$

(10 marks)

Question 1a(ii)

It is known that g is strictly increasing in a neighbourhood of $x = 1$. Use the Inverse Function Theorem to compute the value of $\left(g^{-1}\right)'(6)$.

(4 marks)

Question 1b

Let

$$h(x) = \frac{1}{2 + 3x^2}.$$

Compute the value of $h'(1)$ from first principles, that is, by using the limit definition of the derivative.

(6 marks)

Question 2a

A function $g(x)$ is differentiable on $\mathbb{R}$ and satisfies the property that $g'(x) \geq 2$ for all x. Given that $g(3) = 5$, prove that $g(9) \geq 17$.

(5 marks)

Question 2b

Let m, n be positive integers such that $m \geq 3$ and $5n < 2m^2$, and define a function

$$f(x) = 3x^4 - 4mx^3 - 30nx^2 + 144nx.$$

Question 2b(i)

Show that the function $f(x)$ has a stationary point between $x = m$ and $x = 2m$, that is, a point where the derivative of f vanishes.

(7 marks)

Question 2b(ii)

Given that the stationary point in Question 2b(i) is unique, determine whether it is a local minimum point or a local maximum point.

(3 marks)

Question 2b(iii)

Prove that there exists real numbers a, b with $a < b$ such that f is convex whenever $x < a$ or $x > b$, and f is not convex on the open interval (a, b).

(5 marks)

Question 3

Let n be a positive integer. Suppose that $f : [2.5,\ 3.5] \longrightarrow \mathbb{R}$ is a function that is differentiable 4 times on the closed interval $[2.5,\ 3.5]$, and which satisfies

$$f(3) = 1, f'(3) = -1/3, f''(3) = 4/9, f'''(3) = -28/27, \text{and } f^{(4)}(x) \leq n \text{ for } 2.5 \leq x \leq 3.5.$$

Question 3a

Find the Taylor polynomial $P_3(x)$ of $f(x)$ about the point $x = 3$, that is, the Taylor expansion about $x = 3$ up to and including the term involving $(x-3)^3$.

(4 marks)

Question 3b

Find the maximum error in estimating $f(x)$ in the domain $2.5 \leq x \leq 3.5$ using the Taylor polynomial $P_3(x)$ about the point $x = 3$. Express your answer in terms of n.

(3 marks)

Question 3c

The function

$$g(x) = (x-2)^{-1/3}$$

is infinitely differentiable on $[2.5,\ 3.5]$, and satisfies

$$g(3) = 1, g'(3) = -1/3, g''(3) = 4/9, g'''(3) = -28/27.$$

Question 3c(i)

Determine the smallest possible value of the integer n such that

$$\left|g^{(4)}(x)\right| \leq n$$

for all values of x satisfying $2.5 \leq x \leq 3.5$.

(4 marks)

Question 3c(ii)

Use the Taylor polynomial $P_3(x)$ of $g(x)$ about the point $x = 3$ to estimate the value of $23^{1/3}$. Give your estimate to 4 decimal places.

(4 marks)

Question 4a

Compute the limit

$$\lim_{x \longrightarrow \frac{\pi}{4}} \left(\frac{2\tan(x) - 4x + \pi - 2}{(4x - \pi)^2} \right)$$

using L'Hospital's Rule.

(8 marks)

Question 4b

Compute the limit

$$\lim_{n \longrightarrow \infty} \left[\frac{1}{n^{\frac{1}{4}}} \sum_{i=1}^{n} \left(\frac{1}{(i+n)^{\frac{3}{4}} + n^{\frac{1}{2}}(i+n)^{\frac{1}{4}}} \right) \right]$$

by using an appropriate definite integral.

(8 marks)

Question 4c

Use the Fundamental Theorem of Calculus to evaluate

$$\frac{d}{dx} \left(\int_0^{3x^2+x+1} \cos\left(t^3 + 2\right) dt \right).$$

(4 marks)

Question 5

Define a function

$$f(x) = 6 + x - x^2\,, \quad -2 \le x \le 3\,.$$

A graph of function $y = f(x)$ is provided in Figure Q5 below. The graph $y = f(x)$ intersects the x-axis at the points $(-2, 0)$ and $(3, 0)$. There is an axis of symmetry given by the vertical line $x = 1/2$ which passes through the maximum point $(1/2\,, 25/4)$ on the graph.

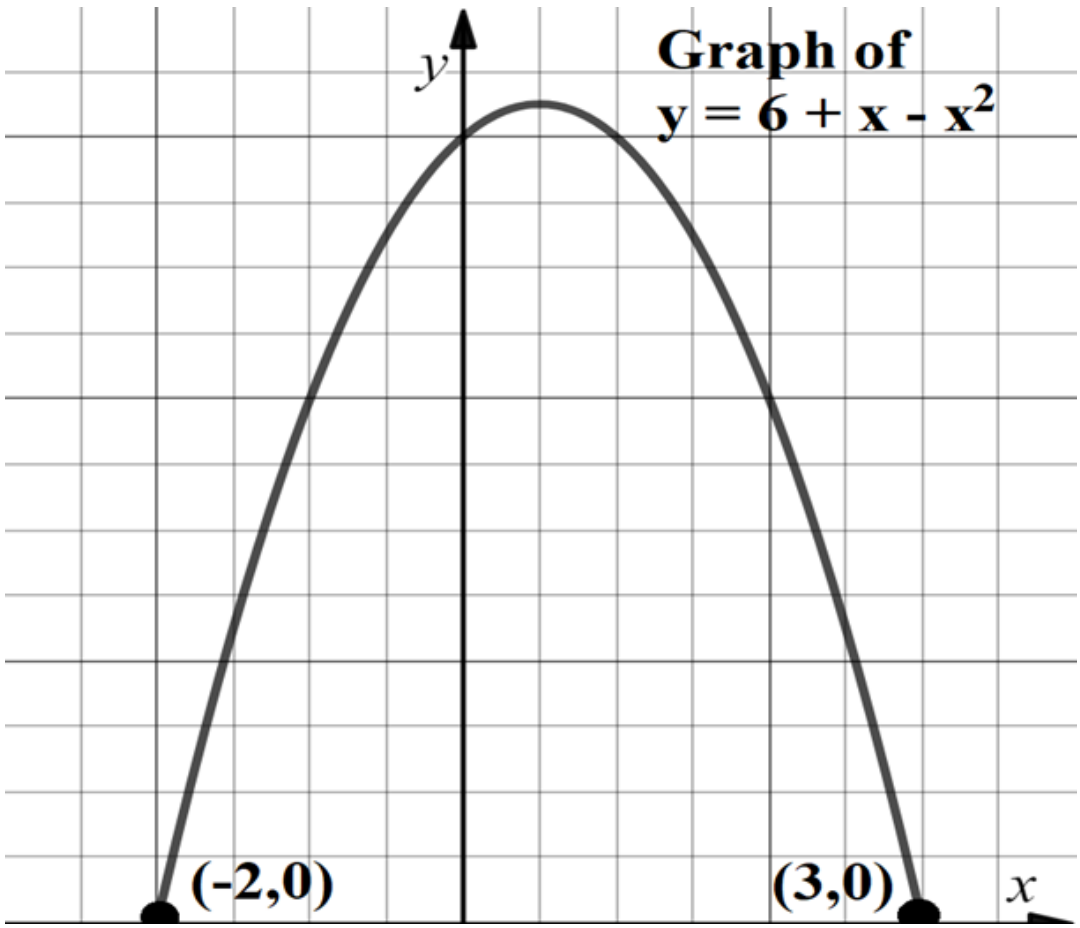

Figure Q5

Question 5a

Let $\mathcal{P}$ denote the tagged partition of $[-2, 3]$ into 7 intervals, consisting of the intervals and tags given below:

$$([-2,\ -1], \text{tag}: -1.5), ([-1,\ -0.5], \text{tag}: -0.6), ([-0.5,\ 0], \text{tag}: -0.3),$$
$$([0,\ 1], \text{tag}: 0.5), ([1,\ 1.5], \text{tag}: 1.3), ([1.5,\ 2], \text{tag}: 1.7), ([2,\ 3], \text{tag}: 2.5).$$

Calculate the value of the Riemann sum $\mathcal{S}(f, \mathcal{P})$. Show full working.

(6 marks)

Question 5b

For any positive integer n, let Q_{2n} denote the partition of $[-2, 3]$ into $2n$ equal-width intervals, namely,

$$\left[-2, -2 + \frac{5}{2n}\right],\ \left[-2 + \frac{5}{2n}, -2 + \frac{2\,(5)}{2n}\right],\ \left[-2 + \frac{2\,(5)}{2n}, -2 + \frac{3\,(5)}{2n}\right],\ \ldots,\ \left[-2 + \frac{(2n-1)(5)}{2n}, 3\right].$$

Show that the Lower Riemann sum $L\,(f, Q_{2n})$ is given by

$$L\,(f, Q_{2n}) = \frac{125}{24n^2}\left(4n^2 + an + b\right),$$

where a and b are integer constants to be determined, and derive a similar expression for the Upper Riemann sum $U(f, Q_{2n})$ in terms of n. Show full working.

(12 marks)

Question 5c

Demonstrate from first principles using the results obtained in Question 5b that f is Riemann integrable on $[-2, 3]$, and that

$$\int_{-2}^{3} f = \frac{125}{6}.$$

(7 marks)

----- END OF PAPER -----